

Jonathan Zhang

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Skills

Graduate Level Coursework: Machine Learning, Causal Inference, Probability, Honors Mathematical Statistics,
Programming: Python, Java, MATLAB, R/R-studio, Flutter/Dart, UNIX/Linux, NumPy, TensorFlow

Experiences

Machine Learning Researcher, Oberst Lab – Baltimore, MD March 2025 - Present

- Developed novel, theoretical bounds on performance of untested ML models under unobserved confounding.
- Formulated empirical, doubly-robust estimator for bounds by demonstrating convergence to true value.
- First-authoring paper on findings with professor and PhD students.

Bioinformatics Researcher, Feinberg Lab – Baltimore, MD May 2025 - Present

- Analyzing whole-genome sequencing data with R and Julia on remote linux cluster.
- Used mediation analysis techniques to statistically rule out confounding effects in existing lab experiments.
- Reduced user input in scripts by over 50% by consolidating disorganized scripts into individual pipelines.
- Sped up runtime by 15% through optimization and error handling of existing data analysis scripts.

COO/Co-founder, Aspire: the Pre-Med Planner – Baltimore, MD June 2024 - Present

- Developed a cross-platform mobile/web app in Flutter/Dart for pre-med students with Firebase backend.
- Created AI-powered chatbot using OpenAI API and engineered prompt to integrate user data into agent context.
- Finalist at SPARK and Hopstart venture competitions, with 500\$ total competition earnings.
- Learn more or download at aspirodoctor.com or Apple App Store

Teaching Assistant, Machine Learning & Probability – Baltimore, MD August 2024 – Present

- Teaches recitation and drafts homework for EN.601.475/675, a graduate level machine learning course.
- Leads a recitation section, holds office hours, and grades student work for EN.553.420/620: Probability, a graduate level probability course.
- Coordinates with professors to plan and develop review material for students.

Immunology Researcher, Baumgarth Lab – Baltimore, MD May 2024 - Dec 2024

- Proficient in handling mice, flow cytometry, cell culture, and western blot data collection/analysis.
- Experienced using R, MATLAB, and ImageJ to analyze and visualize collected data.

Projects

Breast Cancer Identification App for Resident Physicians github.com/jonathanzhang58/MammologyApp

- Collaborating with a JHSOM professor and practicing radiologist to build a training app for over 50 radiology residents in breast cancer identification using IRB approved de-identified patient mammograms
- Developing a publically accessible version; MATLAB App prototype available without patient data.
- Co-authoring a research paper on app impact on residents; Finalist in JHU CBID Healthcare Design Competition

Kalman Filter Visualizer <https://github.com/jonathanzhang58/Kalman-filter-algorithm-and-visualizer>

- Implemented Kalman filter in MATLAB to predict unknown parameters from covariance with measured values.
- Presented findings at departmental conference to over 50 students and professors as part of JHU Directed Reading Program in AMS.

Home Server June 2024

- Transformed old laptop into home server running on Ubuntu Server OS.
- Configured server with VPN to home network enabling remote file storage and video streaming via ssh.
- Packaged aforementioned applications into Docker containers to optimize server for compatibility.

Education

Johns Hopkins University – GPA: 3.9 August 2023 – May 2027

- BS in Applied Maths and Statistics, BS in Biomedical Engineering, Minor in Computer Science